SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: B Tech (Artificial Intelligence, Data Science Computer Engineering, Information Technology, CSE (Cyber), AI and ML, AI and DS, CSBS, CSE (DS), Computer Science) MBA Tech (All Programs)
B Tech Integrated (Computer Engineering)

Year: II/IV Semester: IV/VIII

Stream/s ;CSE-DS/AIDS/CS/AIML/DS/AI/CSBS/COMP/DS

Subject: Database Management System

Time: 03 hrs (10:00 am to 1:00 pm)

Date: 05 / 05 / 2025

No. of Pages:05

Marks: 100

Final Examination 2024-25/Re-Examination (2023-24)/2022-23

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO1 ; SO-3 ; BL-3;	a.	Discuss the role of aggregation in an Extended ER (EER) diagram. How does it help in modeling complex relationships?	[5]
CO-2, SO-1 BL-2	b.	Differentiate between Data Definition Language (DDL) and Data Manipulation Language (DML) in SQL. Provide two commands from each category along with their syntax.	[5]
CO-3, SO-1 BL-2	c.	Explain the Armstrong's axioms with suitable examples.	[5]
CO-4; SO-1 ; BL-3	d.	What are the various states a transaction undergoes during its lifecycle? Explain each state with a well-labeled diagram.	[5]
Q2 CO-1; SO-1; BL-3	a	List the different database users. Briefly state the responsibilities of DBA. What is Data Independence? How does physical data independence differ from logical data independence?	[10]

CO2;SO1 ;BL3	b	Consider the following relational database schema consisting of the four relation schemas: **passenger** (pid, pname, pgender, pcity) **agency** (aid, aname, acity) **flight** (fid, fdate, time, src, dest) **booking** (pid, aid, fid, fdate) Answer the following questions using relational algebra queries. i. Get the details about all flights from Chennai to New Delhi. ii. Find only the flight numbers for passenger with pid 123 for flights to Chennai before 06/11/2020. iii. Find the passenger names for passengers who have bookings on at least one flight. iv. Get the details of flights that are scheduled on both dates 01/12/2020 and 02/12/2020 at 16:00 hours. v. Find the details of all male passengers who are associated with Jet agency.	[10]
Q3 CO- 2, SO-2,3, BL-4	a	A social media platform maintains a database to track users, posts, likes, and comments. The database consists of the following relations: **Tables:** 1. USER (User_ID, Name, Email, Age, Country, Join_Date) 2. POST (Post_ID, User_ID, Content, Post_Date, Likes_Count) 3. COMMENT (Comment_ID, Post_ID, User_ID, Comment_Text, Comment_Date) 4. LIKE (Like_ID, Post_ID, User_ID, Like_Date) Write SQL queries for the following: 1. Create a view that displays the User_ID, Name, Email, and the number of posts made by each user. 2. Retrieve the names and emails of users who have made at least one post. 3. Find the details of posts that have received more likes than the average number of likes on the platform. 4. List all users who have not commented on any post. 5. Create a view that lists all comments along with the post content and the user who made the comment.	[10]
CO3; SO-3 BL: 4	b	What are Data Anomalies in DBMS, and How Do they affect Database Integrity?	[10]

Consider the following relation:

STUDENT_COURSE (StudentID, StudentName, CourseID, CourseName, Instructor, InstructorPhone, Grade).

Functional Dependencies (FDs):

1. StudentID → StudentName
2. CourseID → CourseName, Instructor, InstructorPhone
3. (StudentID, CourseID) → Grade

Normalize the relation up to BCNF.

Q4 CO-1; SO-11; BL-4

a. Design an entity relationship diagram (ERD) for an online super market store management. The super market sells several items. Each item falls into one category. There are several categories of items for example clothes, eatery items, Daily essentials, Grocery, home appliance etc. Each category will have unique category id, category name, category description and category type. Each item will have unique item code, item name, quantity in stock, price, expiry date. The customers can place order for one or more items. The details like order number, order date and total amount will be stored in the database. There are two types of customer who purchases items online from the store, premium customer and non-premium customer. The both type of customer has unique customer id, username, password (to log in into system for purchase), customer name, customer address (for shipping item), and customer phone number. The premium customers are those customers who purchase items costing 10,000 rupees or more every month, hence they are given discount. The discount given will be 10% of total amount. The amount discounted is to be stored in a database. [10]

Q4 CO-2,SO-2, BL-4

b. 1. Describe different types of joins in relational databases and illustrate each with examples.

2.Consider the following two tables representing a store system

OrderDetails

OrderID	BookID	Quantity
0101	B001	5
0102	B003	2
0103	B002	7
0104	B001	1

ClothDetails

BookID	Category
0101	Saree
0102	Suit
0103	Saree

What is the output of the following SQL query?

SELECT Category, SUM (Quantity) FROM OrderDetails, ClothDetails WHERE OrderDetails.BookID=ClothDetails.BookID GROUP BY Category;

[10]

<table>
<tr>
<td>Q5
CO1 ;
SO-2 ;
BL-3;</td>
<td>a</td>
<td>1. Explain the concept of cardinality ratio in an ER diagram. Discuss different types of cardinality ratios with suitable real-world examples.
2. Describe the following terms: a) Strong Entity b) Weak Entity</td>
<td>[10]</td>
</tr>
<tr>
<td>CO3; SO-2 BL: 4</td>
<td>b</td>
<td>Explain the concept of decomposition with an example. Evaluate the conditions for lossless and dependency-preserving decomposition.
Consider the schema R = {A, B, C, D} such that following functional dependency holds on it:

$F = \{A \rightarrow B, A \rightarrow BC, C \rightarrow D\}$

If R is decomposed into relations $\mathbf{R_1} = \{A, B\}$ and $\mathbf{R_2} = \{B, C, D\}$, check if decomposition results in good design or not.</td>
<td>[10]</td>
</tr>
<tr>
<td>Q6
CO-2;
SO-2;
BL-3</td>
<td>a</td>
<td>1. Consider the following tables
<table>
<tr><th colspan="3">CityInfo</th></tr>
<tr><td>CityName</td><td>Population</td><td>StateName</td></tr>
<tr><td>Mumbai</td><td>20000</td><td>Maharashtra</td></tr>
<tr><td>Delhi</td><td>19000</td><td>Delhi</td></tr>
<tr><td>Bengaluru</td><td>12000</td><td>Karnataka</td></tr>
<tr><td>Hyderabad</td><td>10000</td><td>Telangana</td></tr>
<tr><td>Ahmadabad</td><td>8000</td><td>Gujrat</td></tr>
<tr><td>Pune</td><td>6000</td><td>Maharashtra</td></tr>
</table>
Write the relational algebra query to produce the following output from the above table.
<table>
<tr><td>CityName</td><td>Population</td><td>StateName</td></tr>
<tr><td>Mumbai</td><td>20000</td><td>Maharashtra</td></tr>
<tr><td>Delhi</td><td>19000</td><td>Delhi</td></tr>
<tr><td>Bengaluru</td><td>12000</td><td>Karnataka</td></tr>
</table>
2 .Describe the role of Primary and Foreign Keys in relational databases, how they enforce uniqueness and referential integrity, and provide an example scenario where they are used together.</td>
<td>[10]</td>
</tr>
<tr>
<td>CO-4;
SO-2;
BL-4</td>
<td>b</td>
<td>1 Why concurrency control is needed. Explain with examples.
2 Consider the following schedule S involving five transactions T1, T2, T3, T4, and T5:</td>
<td>[10]</td>
</tr>
</table>

T_1	T_2	T_3	T_4	T_5
	W(X)			
R(X)				
W(Y)				
		W(X)		
	R(Z)			
			W(Y)	
				R(X)
				W(Z)

R(X) denotes read operation on data item X by transaction Ti.
W(X) denotes write operation on data item X by transaction Ti
Check whether the given Schedule S is conflict serializable or view serializable or both.

Q7
CO4; SO-1 BL: 2

a. What are the four main types of NoSQL databases? Explain each type with examples. Explain the characteristics of MongoDB [10]

CO4; SO-2 BL: 4

b. Consider the following schedule S involving five transactions T1, T2, T3, T4 and T5: [10]

T_1	T_2	T_3	T_4	T_5
R(Z)				
R(X)				
		R(Y)		
	W(X)			
		W(X)		
			W(Z)	
				W(Z)

R(X) denotes read operation on data item X by transaction Ti
W(X) denotes write operation on data item X by transaction Ti
Check whether the given Schedule S is conflict serializable or view serializable.
Identify which of the following order of executions of all transaction in a schedule is incorrect.

a) T1→T4→T5→T2→T3
b) T1→T2→T3→T4→T5
c) T1→T2→T4→T5→T3
d) T1→T4→T3→T5→T2